



Selecting The Right Motor For Your Design

With so many kinds of motors available in the market, often it becomes difficult to select the most appropriate for an application. So, in such situations, the designers go for anything that comes to their mind as a suitable motor—without looking at the other options that could be better. Here is an introduction to all the motors that are available to choose from, and depending on the application designers can now select the most suitable one for an application

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Here are some guidelines for selecting the best motor for an application, among the many available in the market, such as the universal motor (UM), AC induction motor (IM), permanent magnet motor (PMM), permanent magnet synchronous motor (PMSM), synchronous reluctance motor (SynRM), switched reluctance motor (SRM), and the latest assistive permanent magnet synchronous reluctance motor (IPM-SynRM) offered by Tesla Motors.

Engineers have many options to choose from for their projects. Picking the wrong motor can condemn the design to normality and market failure.

Universal motors

The universal motor (UM) is a type of electric motor that can run on AC or DC power and uses an electromagnet as a stator to create its magnetic field. It is a switched series-wound motor where the stator field coils are

connected in series with the rotor windings through a commutator. It is often referred to as the AC series engine.

The universal motor is very similar to a DC series motor in construction but is slightly modified to allow the motor to operate correctly on AC power. This type of electric motor can operate well on AC because the

current in the field coils and armature (and the resulting magnetic fields) will alternate (reverse polarity) in sync with the supply. Consequently, the resulting mechanical force will occur in a consistent direction of rotation, independent of the direction of the applied voltage, but determined by the commutator and polarity of the field coils.

Universal motors have high starting torque, can run at high speed, and are light and compact. These are commonly used in portable electrical tools and equipment, as well as in many household appliances. The motors are relatively easy to control, electromechanically using derived coils or electronically. However, the commutator has brushes that wear out, so these are much less used for equipment that is in continuous use. Also, in part because of the commutator, universal motors are often very noisy, both acoustically and electromagnetically.

Not all series-wound motors work well on AC current. If a DC motor with a common series coil were connected to an AC source, it would malfunction. The universal motor is modified in several ways to allow for proper operation of AC power. There is a normally added compensating winding, along with the laminated pole pieces, as opposed to the solid pole pieces found in DC motors. The armature of a universal motor typically has many more coils and plates than a DC motor and therefore fewer windings per coil. This reduces the inductance.

Even when used with AC power, these motors can run at a rotational frequency well above the mains supply and, as most electric motor properties improve with speed, this means they can be light and powerful. However, universal motors are generally relatively



Fig. 1: A universal motor